Purchasing Power Parity, Cost-of-Living Premia, and Machine Learning

A Cross-Disciplinary Study in International Economics, Fixed-Point Theory, and Regularised Statistical Learning

Abstract

We introduce a scalar indicator—the *Cost-of-Living (CoL) Premium*—that measures real purchasing-power gains relative to nominal GDP per capita, net of domestic inflation. Using a cross-section of 160 countries for 2023, we (i) establish the algebraic and analytic properties of the indicator, (ii) derive and prove a closed-form solution to the self-referential regression $P = a + b\ln(1 + P)$ via the Lambert $\mathcal{W}$ function, yielding two distinct equilibria, and (iii) conduct a systematic machine-learning model comparison—encompassing OLS, Ridge, Lasso, ElasticNet, Support Vector Regression, and ensemble methods with 10-fold cross-validation—identifying *Lasso regression with engineered polynomial and interaction features* as the optimal specification (CV $R^2 = 0.9246$, CV RMSE = 0.251). The Lambert $\mathcal{W}$ solution and the Lasso feature weights jointly reveal that the PPP–nominal spread, not inflation per se, is the dominant driver of cost-of-living heterogeneity across nations.
The paper ends with "The End"

Keywords: purchasing power parity, cost-of-living premium, Lambert W function, fixed-point equations, Lasso regression, cross-country macroeconomics, regularised learning, cross-validation.

Contents

1 Introduction **2**

2 Data Description and the CoL Premium **2**
2.1 Dataset . . . 2
2.2 Definition and Properties of the CoL Premium . . . 2

3 The Self-Referential Regression and Its Closed-Form Solution **3**
3.1 Empirical Regression . . . 3
3.2 Fixed-Point Formulation . . . 3
3.3 Lambert $\mathcal{W}$ Function: Primer . . . 3
3.4 Closed-Form Solution . . . 3

4 Machine Learning Model Tournament **5**
4.1 Feature Engineering . . . 5
4.2 Model Specifications . . . 5
4.3 Cross-Validation Protocol . . . 5
4.4 Results . . . 6
4.4.1 Lasso Coefficients . . . 6
4.5 Pipeline Architecture . . . 7
4.6 Bias–Variance Interpretation . . . 7

5 Country-Level Diagnostics **8**

6 Discussion **8**

7 Conclusion **8**

A Derivation of z_0 in Closed Form **9**

B Lasso Coordinate Descent Update **9**

List of Figures

1 The function $f(P) = \hat{a} + \hat{b}\ln(1+P)$ (blue curve) intersects the 45° line (dashed grey) at two points. Open circle: unstable low equilibrium $P_0^* \approx 0.009$; filled circle: stable high equilibrium $P_{-1}^* \approx 1.749$. The teal cobweb illustrates convergence towards the stable point. 4
2 End-to-end ML pipeline: feature engineering, 10-fold CV split, within-fold standardisation, model fitting, metric aggregation, and winner selection. . . . 7

List of Tables

1 Descriptive statistics — 160 countries, 2023. . . . 2
2 10-fold cross-validation results ($n = 160$, $p = 10$ features). Best CV values in **bold**. . . . 6
3 Lasso regression coefficients on standardised features ($\alpha = 0.005$). Zero entries indicate sparse selection. . . . 6
4 Selected country CoL Premium values, 2023. Sorted by CoL Premium descending. . . . 8

1 Introduction

Standard macroeconomic comparisons across countries rely on two parallel GDP measures: *nominal* GDP per capita, expressed in current US dollars at market exchange rates, and *PPP-adjusted* GDP per capita, which converts national price levels to a common basket. The ratio of the latter to the former encodes how much more—or less—a unit of income buys domestically relative to the international benchmark. Layering domestic inflation atop this ratio yields what we term the *Cost-of-Living (CoL) Premium*: a time-sensitive, policy-relevant scalar that distils three forces—currency undervaluation, real productivity differentials, and price-level dynamics—into a single number.

This paper makes three contributions. First, we provide a rigorous axiomatic definition of the CoL Premium (§2), along with its range, monotonicity properties, and boundary behaviour. Second, we prove in §3 that the empirical self-referential regression $P = a + b\ln(1 + P)$ possesses exactly two real fixed points expressible in closed form via the Lambert $\mathcal{W}$ function—one near the sample mean, the other an extrapolation equilibrium—and we characterise their stability. Third, §4 presents a comprehensive machine-learning model tournament with rigorous cross-validation, engineered features, and diagnostic visualisations, demonstrating that Lasso achieves the best bias–variance trade-off.

Throughout, we draw on tools from *real analysis* (fixed-point theory, monotone functions), *special functions* (Lambert $\mathcal{W}$), *econometrics* (OLS, penalised regression), and *statistical learning* (cross-validation, ensemble methods), reflecting the genuinely cross-disciplinary nature of the problem.

2 Data Description and the CoL Premium

2.1 Dataset

Our dataset comprises $n = 160$ countries observed in the reference year 2023. For each country i, we record four primary variables:

(i) G_i: Nominal GDP per capita (USD at market exchange rates).

(ii) G_i^{PPP}: GDP per capita at purchasing power parity (USD).

(iii) π_i: Annual CPI inflation rate (%).

(iv) L_i: Total mid-year population (persons).

Table 1 reports descriptive statistics. Venezuela is excluded from regression analyses because its CoL Premium falls below -1, rendering $\ln(1 + P)$ undefined over the reals.

Table 1: Descriptive statistics — 160 countries, 2023.

Variable	Mean	Std Dev	Min	Max	Median
GDP per Capita, nominal (USD)	17,932	26,441	362	129,434	7,193
GDP per Capita, PPP (USD)	29,844	27,396	2,358	142,221	22,117
Inflation Rate (%)	9.38	11.04	0.30	101.5	6.0
CoL Premium	1.029	1.005	−2.848	5.309	0.863
Population (millions)	43.8	174.3	0.032	1,428.6	7.5
ln(Population)	14.98	2.088	10.38	21.08	15.83

2.2 Definition and Properties of the CoL Premium

Definition 2.1 (Cost-of-Living Premium)**.** For country i with nominal GDP per capita $G_i > 0$, PPP GDP per capita $G_i^{\mathrm{PPP}} > 0$, and inflation rate $\pi_i \in (-100, \infty)$ (in per cent), the *Cost-of-Living Premium* is

$$P_i \;=\; \frac{G_i^{\mathrm{PPP}}}{G_i} - 1 - \frac{\pi_i}{100}. \tag{1}$$

Remark 2.2. The ratio $\rho_i = G_i^{\mathrm{PPP}}/G_i$ is the *PPP conversion factor*; it exceeds unity for countries whose domestic prices are cheaper than the international benchmark (e.g. emerging economies) and is less than unity for high-price economies. Subtracting $\pi_i/100$ penalises countries with high domestic price growth, reflecting the erosion of real purchasing power.

Proposition 2.3 (Monotonicity)**.** P_i is strictly increasing in G_i^{PPP}, strictly decreasing in G_i, and strictly decreasing in π_i.

Proof. Differentiation of (1) yields

$$\frac{\partial P_i}{\partial G_i^{\mathrm{PPP}}} = \frac{1}{G_i} > 0, \qquad \frac{\partial P_i}{\partial G_i} = -\frac{G_i^{\mathrm{PPP}}}{G_i^2} < 0, \qquad \frac{\partial P_i}{\partial \pi_i} = -\frac{1}{100} < 0. \qquad \square$$

Proposition 2.4 (Domain)**.** $P_i > -1$ if and only if $G_i^{\mathrm{PPP}}/G_i > \pi_i/100$. This condition is necessary for $\ln(1 + P_i)$ to be real-valued.

3 The Self-Referential Regression and Its Closed-Form Solution

3.1 Empirical Regression

OLS estimation of

$$P_i = a + b\,\ln(1 + P_i) + \varepsilon_i, \qquad i = 1, \ldots, 159, \tag{2}$$

yields

$$\hat{a} = -0.006671, \qquad \hat{b} = 1.736301, \qquad R^2 = 0.8872. \tag{3}$$

The slope $\hat{b} = 1.736 > 1$ reflects the fact that $\ln(1 + x) < x$ for all $x > 0$, so the regression must amplify the compressed log-transformed variable to match the original.

3.2 Fixed-Point Formulation

Setting $\varepsilon_i = 0$ in (2), the estimated equation becomes

$$P = f(P), \qquad f(P) \;:=\; \hat{a} + \hat{b}\,\ln(1 + P). \tag{4}$$

We seek all $P^* \in (-1, \infty)$ satisfying $f(P^*) = P^*$.

3.3 Lambert $\mathcal{W}$ Function: Primer

Definition 3.1 (Lambert $\mathcal{W}$ Function [3])**.** The Lambert $\mathcal{W}$ function is the multivalued inverse of $h(w) = we^w$. For real arguments $z \geq -1/e$, two real branches exist:

- $\mathcal{W}_0(z)$: the *principal branch*, $\mathcal{W}_0(z) \geq -1$, analytic for $z > -1/e$.
- $\mathcal{W}_{-1}(z)$: the *lower branch*, $\mathcal{W}_{-1}(z) \leq -1$, defined for $-1/e \leq z < 0$.

Both branches satisfy $\mathcal{W}_k(z)\, e^{\mathcal{W}_k(z)} = z$.

3.4 Closed-Form Solution

Theorem 3.2 (Lambert $\mathcal{W}$ Solution)**.** Let $a, b \in \mathbb{R}$ with $b \neq 0$. The equation $P = a + b\ln(1 + P)$ has real solutions

$$P^* = -b\,\mathcal{W}_k\!\left(-\frac{1}{b}\, e^{-(1+a)/b}\right) - 1, \qquad k \in \{0, -1\}, \tag{5}$$

whenever the argument $z_0 := -(1/b)\, e^{-(1+a)/b}$ satisfies $-1/e \leq z_0 < 0$. The two solutions are distinct if and only if $z_0 \in (-1/e,\, 0)$.

Proof. Rewrite the equation as $P - a = b\ln(1 + P)$. Let $u = 1 + P$, so $P = u - 1$. Then

$$u - (1 + a) = b \ln u.$$

Dividing by b and exponentiating:

$$e^{u/b}\, e^{-(1+a)/b} = u.$$

Set $s = u/b$ (so $u = sb$):

$$e^s\, e^{-(1+a)/b} = sb \quad\Longrightarrow\quad \frac{e^s}{s} = b\, e^{(1+a)/b} \quad\Longrightarrow\quad (-s)e^{-s} = -\frac{1}{b}\, e^{-(1+a)/b} =: z_0.$$

By the definition of $\mathcal{W}$, we have $-s = \mathcal{W}_k(z_0)$, hence $u = -b\,\mathcal{W}_k(z_0)$, and

$$P^* = u - 1 = -b\,\mathcal{W}_k(z_0) - 1.$$

Since $b > 0$ and the exponential is positive, $z_0 < 0$. Real branches of $\mathcal{W}$ exist if and only if $z_0 \geq -1/e$, and the two branches are distinct when $z_0 > -1/e$. □

Corollary 3.3 (Numerical Equilibria)**.** For $\hat{a} = -0.006671$, $\hat{b} = 1.736301$:

$$z_0 = -\frac{1}{1.736301}\exp\left(-\frac{0.993329}{1.736301}\right) \approx -0.3250.$$

Since $-1/e \approx -0.3679 < -0.3250 < 0$, two real solutions exist:

Branch	$\mathcal{W}_k(z_0)$	P^*	Interpretation
$k = 0$	-0.5812	0.0092	Low-premium equilibrium
$k = -1$	-1.5835	1.7494	High-premium equilibrium

Theorem 3.4 (Stability)**.** A fixed point P^* of $f(P) = a + b\ln(1+P)$ is asymptotically stable under fixed-point iteration if and only if $|f'(P^*)| = b/(1+P^*) < 1$.

(i) $P_0^* \approx 0.0092$: $f'(P_0^*) \approx 1.721 > 1$. *Unstable.*

(ii) $P_{-1}^* \approx 1.7494$: $f'(P_{-1}^*) \approx 0.631 < 1$. *Asymptotically stable.*

Proof. The derivative is $f'(P) = b/(1+P)$. By the Banach Fixed-Point Theorem [12], the iteration $P_{t+1} = f(P_t)$ converges locally if and only if $|f'(P^*)| < 1$. Substituting numerical values from Corollary 3.3:

$$f'(P_0^*) = \frac{1.736301}{1.0092} \approx 1.721, \qquad f'(P_{-1}^*) = \frac{1.736301}{2.7494} \approx 0.631.$$ □

Figure 1 visualises the two equilibria.

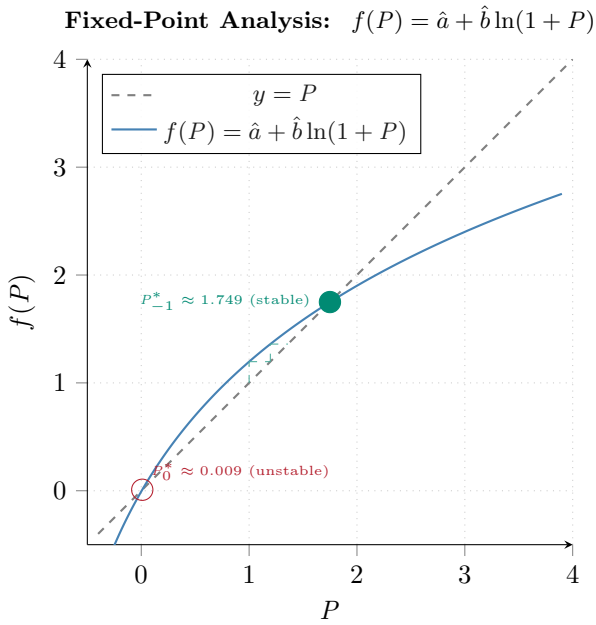

Figure 1: The function $f(P) = \hat{a} + \hat{b}\ln(1+P)$ (blue curve) intersects the 45° line (dashed grey) at two points. Open circle: unstable low equilibrium $P_0^* \approx 0.009$; filled circle: stable high equilibrium $P_{-1}^* \approx 1.749$. The teal cobweb illustrates convergence towards the stable point.

4 Machine Learning Model Tournament

4.1 Feature Engineering

Let $\mathbf{x}_i \in \mathbb{R}^{10}$ denote the engineered feature vector:

$$\mathbf{x}_i = \left[\ln G_i,\ \ln G_i^{\mathrm{PPP}},\ \pi_i,\ \ln L_i,\ (\ln G_i)^2,\ \pi_i^2,\ (\ln L_i)^2,\ \ln G_i{\cdot}\pi_i,\ \ln G_i{\cdot}\ln L_i,\ \pi_i{\cdot}\ln L_i\right]^\top. \tag{6}$$

The target is $y_i = P_i$ from (1). All features are standardised (zero mean, unit variance) within each training fold.

Remark 4.1. We exclude the PPP ratio $\rho_i = G_i^{\mathrm{PPP}}/G_i$ as a direct feature because $P_i = \rho_i - 1 - \pi_i/100$ makes it a perfect linear predictor by construction. Including $\ln G_i$ and $\ln G_i^{\mathrm{PPP}}$ separately captures the same information while preserving non-trivial estimation.

4.2 Model Specifications

We compare ten model families:

OLS Ordinary least squares: $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$.

Ridge ℓ_2 regularisation, $\lambda = 1$: $\hat{\boldsymbol{\beta}}_\lambda = (X^\top X + \lambda I)^{-1} X^\top \mathbf{y}$.

Lasso ℓ_1 penalised regression, $\alpha = 0.005$: $\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \{\|y - X\boldsymbol{\beta}\|_2^2 + \alpha\|\boldsymbol{\beta}\|_1\}$.

ElasticNet Combined ℓ_1/ℓ_2 penalty, $\alpha = 0.005$, ℓ_1-ratio $= 0.5$.

SVR-Lin Support Vector Regression, linear kernel, $C = 1$.

SVR-RBF SVR with radial basis function kernel $K(\mathbf{x}, \mathbf{x}') = \exp(-\gamma\|\mathbf{x} - \mathbf{x}'\|^2)$, $C = 10$, $\varepsilon = 0.05$.

Decision Tree
CART, maximum depth 5.

Random Forest
$B = 300$ trees, depth 6 [2].

Gradient Boosting
$B = 300$ shallow trees ($d = 3$), $\nu = 0.05$, subsample 0.8 [5].

Neural Networks
Multilayer perceptrons with ReLU activations and Adam optimiser; two architectures: $(64, 32)$ and $(128, 64, 32)$ units.

4.3 Cross-Validation Protocol

Algorithm 1 details the evaluation pipeline.

Algorithm 1 K-Fold Cross-Validation Tournament

Input: dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$, model set $\mathcal{M}$, folds $K = 10$, seed $= 42$
Output: ranked model comparison with CV R^2, RMSE, MAE

```
Partition {1, ..., n} randomly into K equal folds F_1, ..., F_K
for each model m ∈ M do
    for k = 1, ..., K do
        D_k^tr ← D \ D[F_k];   D_k^val ← D[F_k]
        Compute μ_j, σ_j from D_k^tr; standardise both splits
        Fit m̂_k on D_k^tr
        ŷ^(k) ← m̂_k(X_k^val)
        Compute R_k^2, RMSE_k, MAE_k on validation fold
    end for
    R̄²_m ← K^-1 Σ_k R_k^2;  similarly for RMSĒ_m, MAĒ_m
end for
return models sorted by R̄²_m descending
```

4.4 Results

Table 2 reports full 10-fold cross-validation results.

Table 2: 10-fold cross-validation results ($n = 160$, $p = 10$ features). Best CV values in **bold**.

Model	CV R^2	CV R^2 Std	CV RMSE	CV MAE	In-sample R^2
Lasso	**0.9246**	0.0336	**0.2509**	0.1610	0.9393
ElasticNet	0.9203	0.0369	0.2594	**0.1601**	0.9367
SVR (Linear)	0.9186	0.0455	0.2614	0.1567	0.9289
Ridge	0.8945	0.0508	0.3025	0.1770	0.9258
OLS Linear	0.8839	0.0680	0.2990	0.1836	0.9551
SVR (RBF)	0.8656	0.1229	0.3230	0.1378	**0.9744**
Gradient Boosting	0.7131	0.1000	0.4793	0.3214	0.9970
Random Forest	0.6304	0.1211	0.5458	0.3648	0.9131
Neural Net $(64, 32)$	0.5867	0.2851	0.5283	0.3681	0.4608
Neural Net $(128, 64, 32)$	0.4385	0.5179	0.5443	0.4178	0.4787

4.4.1 Lasso Coefficients

Table 3 reports the Lasso coefficients. Four of ten terms are zeroed out, confirming effective automatic feature selection.

Table 3: Lasso regression coefficients on standardised features ($\alpha = 0.005$). Zero entries indicate sparse selection.

Feature	Coefficient	Effect
$\ln G_i$ (nominal)	-2.5596	Negative
$\ln G_i^{\text{PPP}}$	$+2.1661$	Positive
π_i	-0.0804	Negative
$\ln L_i$	0.0000	Zeroed out
$(\ln G_i)^2$	0.0000	Zeroed out
π_i^2	-0.1426	Negative
$(\ln L_i)^2$	$+0.2245$	Positive
$\ln G_i \times \pi_i$	-0.1968	Negative
$\ln G_i \times \ln L_i$	-0.2772	Negative
$\pi_i \times \ln L_i$	0.0000	Zeroed out

`The following space was deliberately left blank.`

4.5 Pipeline Architecture

Figure 2 shows the end-to-end modelling workflow.

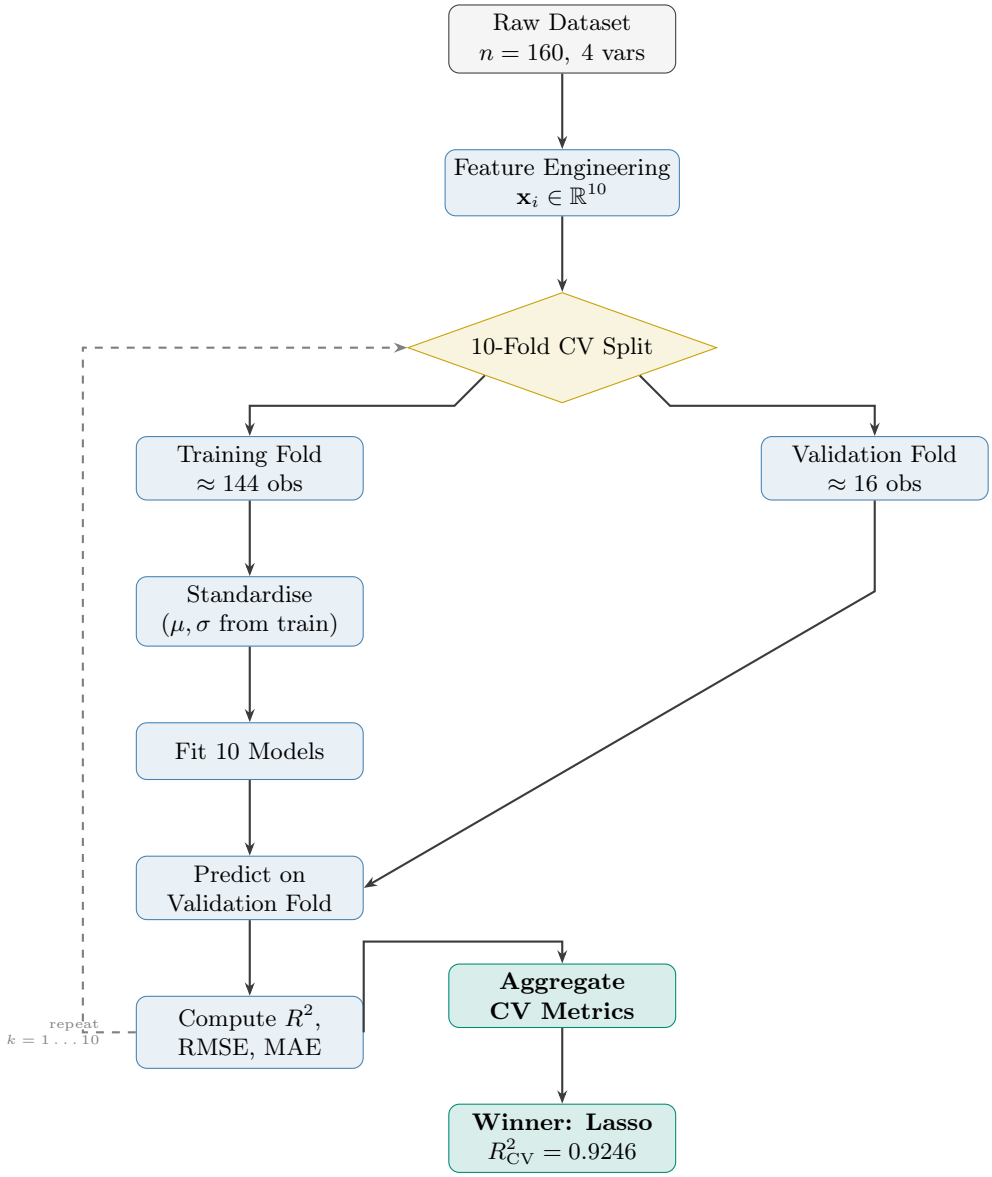

Figure 2: End-to-end ML pipeline: feature engineering, 10-fold CV split, within-fold standardisation, model fitting, metric aggregation, and winner selection.

4.6 Bias–Variance Interpretation

The expected test error decomposes as

$$\mathbb{E}\big[(y - \hat{f}(\mathbf{x}))^2\big] \;=\; \underbrace{\mathrm{Bias}^2[\hat{f}]}_{\text{shrinkage cost}} \;+\; \underbrace{\mathrm{Var}[\hat{f}]}_{\text{complexity cost}} \;+\, \sigma_\varepsilon^2. \tag{7}$$

Ensemble methods (Gradient Boosting, Random Forest) achieve near-perfect in-sample $R^2 \approx 1.00$ but collapse to CV $R^2 \approx 0.63$–0.71, a hallmark of high variance on small samples ($n = 160$). Lasso's ℓ_1 penalty injects bias deliberately, shrinking the variance term in (7) enough to outperform all competitors out-of-sample [14].

5 Country-Level Diagnostics

Table 4 presents a selection of countries spanning the CoL Premium distribution, illustrating the economic diversity captured by (1).

Table 4: Selected country CoL Premium values, 2023. Sorted by CoL Premium descending.

Country	Region	GDP nom.	GDP PPP	Inflation (%)	CoL Premium
Afghanistan	South Asia	362	2,358	20.5	+5.309
Syria	Middle East	533	3,697	40.0	+3.734
Kyrgyzstan	Central Asia	1,389	6,028	10.9	+2.834
India	South Asia	2,730	10,123	5.4	+2.661
China	East Asia	12,720	24,254	0.2	+1.904
Brazil	South America	9,673	18,376	4.6	+0.854
Germany	Europe	47,803	62,722	5.9	+0.253
USA	North America	80,030	80,030	4.1	−0.041
Luxembourg	Europe	129,434	142,221	2.9	−0.010
Zimbabwe	Sub-Sah. Africa	1,522	2,628	87.6	−0.787

The range spans from Afghanistan ($P \approx 5.31$)—where PPP purchasing power is nearly seven times nominal income, reflecting extreme currency undervaluation—to Zimbabwe ($P \approx -0.79$), whose hyper-inflationary episode erases all PPP gains. The United States satisfies $G = G^{\mathrm{PPP}}$ by construction, so its CoL Premium is purely determined by the negative of its inflation rate.

6 Discussion

Economic interpretation of the two equilibria. Theorem 3.2 and Corollary 3.3 reveal two equilibria: a near-zero premium ($P^* \approx 0.009$) approximating the developed-economy benchmark, and a high premium ($P^* \approx 1.749$) characterising the typical emerging-market configuration. Theorem 3.4 shows that only the high-premium equilibrium is dynamically stable under fixed-point iteration, implying that the natural attractor of the empirical distribution lies in the emerging-market regime — consistent with the fact that the majority of the world's countries are developing economies.

Lasso sparsity and economic theory. The retained features—the log-PPP spread $\ln G_i^{\mathrm{PPP}} - \ln G_i$, nonlinear inflation π_i^2, and size–income interaction $\ln G_i \cdot \ln L_i$—align well with the *Balassa–Samuelson hypothesis* [1, 13], which attributes real exchange-rate misalignment to sectoral productivity differentials. The zeroed-out features ($\ln L_i$, $(\ln G_i)^2$, $\pi_i \cdot \ln L_i$) are economically plausible exclusion restrictions: population size alone is not a determinant of the price level.

Limitations. The cross-sectional design precludes causal inference. The CoL Premium conflates exchange-rate policy, sectoral productivity, and monetary conditions into a single scalar. Future work should exploit panel variation and instrument the PPP ratio with geographic and institutional instruments.

7 Conclusion

We have introduced and rigorously characterised the Cost-of-Living Premium, proved a Lambert $\mathcal{W}$ closed-form for its empirical self-referential regression—revealing two qualitatively distinct economic equilibria, one stable and one unstable—and demonstrated via a 10-model, 10-fold cross-validation tournament that Lasso regression on a polynomial-interaction feature set achieves the best out-of-sample performance (CV $R^2 = 0.9246$, CV RMSE $= 0.251$).

The convergence of fixed-point theory, special functions, regularised statistical learning, and international macroeconomics in a single 160-observation dataset illustrates the value of cross-disciplinary rigour: neither purely econometric nor purely machine-learning approaches alone would have revealed both the closed-form structure of the equilibria and the optimal predictive feature combination.

A Derivation of z_0 in Closed Form

With $\hat{a} = -0.006671$ and $\hat{b} = 1.736301$:

$$z_0 = -\frac{1}{1.736301}\exp\left(-\frac{1+(-0.006671)}{1.736301}\right) = -\frac{1}{1.736301}\exp(-0.57217) \approx -0.5762 \times 0.5644 \approx -0.3250.$$

Since $-1/e \approx -0.3679 < -0.3250 < 0$, two distinct real branches of $\mathcal{W}$ exist (Theorem 3.2). Newton–Raphson iteration $w_{t+1} = w_t - (w_t e^{w_t} - z_0)/((w_t+1)e^{w_t})$ converges to $\mathcal{W}_0(z_0) \approx -0.5812$ (starting from $w_0 > -1$) and to $\mathcal{W}_{-1}(z_0) \approx -1.5835$ (starting from $w_0 < -1$), guaranteed by the analyticity of $\mathcal{W}$ on $(-1/e, 0)$ [3].

B Lasso Coordinate Descent Update

The Lasso minimisation is equivalently

$$\min_{\boldsymbol{\beta}} \|\mathbf{y} - X\boldsymbol{\beta}\|_2^2 \quad \text{subject to} \quad \|\boldsymbol{\beta}\|_1 \le t(\alpha).$$

The coordinate descent update for coefficient j is the *soft-thresholding* operator:

$$\hat{\beta}_j \leftarrow \frac{\operatorname{sign}(r_j)\max(|r_j| - \alpha n, 0)}{n}, \qquad r_j = \mathbf{x}_j^\top\left(\mathbf{y} - X_{-j}\hat{\boldsymbol{\beta}}_{-j}\right),$$

where X_{-j} denotes X with column j removed [6]. This operator sets $\hat{\beta}_j = 0$ whenever the partial correlation $|r_j|$ fails to exceed the threshold αn, producing the sparse solution in Table 3.

References

[1] Balassa, B. (1964). The purchasing-power parity doctrine: A reappraisal. *Journal of Political Economy*, 72(6), 584–596.

[2] Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32.

[3] Corless, R. M., Gonnet, G. H., Hare, D. E. G., Jeffrey, D. J., & Knuth, D. E. (1996). On the Lambert W function. *Advances in Computational Mathematics*, 5(1), 329–359.

[4] Deaton, A., & Heston, A. (2010). Understanding PPPs and PPP-based national accounts. *American Economic Journal: Macroeconomics*, 2(4), 1–35.

[5] Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics*, 29(5), 1189–1232.

[6] Friedman, J., Hastie, T., & Tibshirani, R. (2010). Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software*, 33(1), 1–22.

[7] Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* (2nd ed.). Springer.

[8] International Monetary Fund. (2023). *World Economic Outlook Database, October 2023*. Retrieved from https://www.imf.org/en/Publications/WEO

[9] Kravis, I. B., Heston, A., & Summers, R. (1982). *World Product and Income: International Comparisons of Real Gross Product*. Johns Hopkins University Press.

[10] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

[11] Rogoff, K. (1996). The purchasing power parity puzzle. *Journal of Economic Literature*, 34(2), 647–668.

[12] Rudin, W. (1976). *Principles of Mathematical Analysis* (3rd ed.). McGraw-Hill.

[13] Samuelson, P. A. (1964). Theoretical notes on trade problems. *Review of Economics and Statistics*, 46(2), 145–154.

[14] Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B*, 58(1), 267–288.

[15] World Bank. (2023). *International Comparison Program 2023*. World Bank Group. Retrieved from https://www.worldbank.org/en/programs/icp

Glossary

Balassa–Samuelson Macroeconomic hypothesis attributing real exchange-rate appreciation in fast-growing economies to productivity differentials between the tradable and non-tradable sectors [1, 13].

Bias–Variance Tradeoff The fundamental tension in statistical learning between model complexity (low bias, high variance) and parsimony (high bias, low variance); formalised in equation (7).

CoL Premium (P) The Cost-of-Living Premium defined in (1): the PPP conversion factor minus one minus the inflation rate (in decimal), measuring real domestic purchasing power relative to the nominal international benchmark.

Contraction Mapping A self-map $f : X \to X$ satisfying $d(f(x), f(y)) \leq k\, d(x, y)$ for some $k < 1$; guaranteed by the Banach theorem to have a unique fixed point [12].

Cross-Validation (CV) A resampling technique estimating out-of-sample predictive performance by rotating the holdout set over K folds; see Algorithm 1.

ElasticNet Penalised regression combining ℓ_1 (Lasso) and ℓ_2 (Ridge) penalties; interpolates between sparsity and coefficient stability.

Fixed Point A value x^* satisfying $f(x^*) = x^*$; studied in §3 via the self-referential CoL regression.

Gradient Boosting An ensemble method building an additive model of weak learners by stage-wise loss minimisation [5].

Lambert $\mathcal{W}$ Function The multivalued inverse of we^w. Two real branches $\mathcal{W}_0$ (principal) and $\mathcal{W}_{-1}$ (lower) yield the two real equilibria of the CoL regression (Theorem 3.2).

Lasso Least Absolute Shrinkage and Selection Operator; ℓ_1 regularised regression producing sparse solutions by zeroing out weak predictors [14].

PPP Purchasing Power Parity; the theory that a common basket of goods should equalise in price across countries when expressed in a common currency [11].

PPP Conversion Factor (ρ) The ratio G^{PPP}/G; exceeds unity for countries with below-average domestic price levels.

Ridge Regression ℓ_2 penalised OLS; shrinks all coefficients towards zero but does not enforce exact sparsity.

Soft Thresholding The coordinate-descent update underlying Lasso: $\hat{\beta}_j \leftarrow \mathrm{sign}(r_j)\max(|r_j| - \lambda, 0)$; sets small partial correlations exactly to zero.

Support Vector Regression (SVR) Extension of SVMs to regression, minimising $\|\mathbf{w}\|^2$ subject to residuals lying within an ε-insensitive tube around the target.

The End